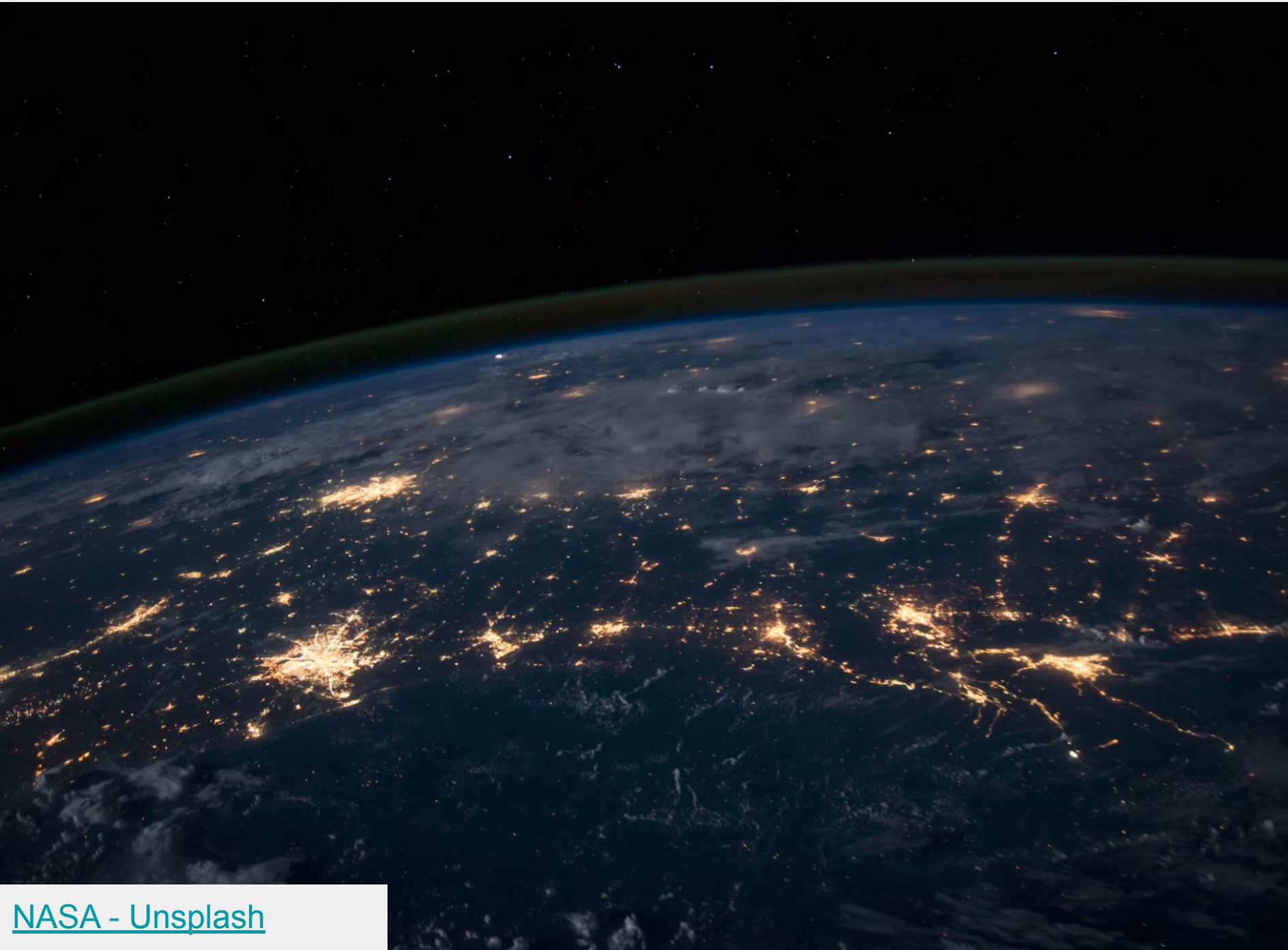


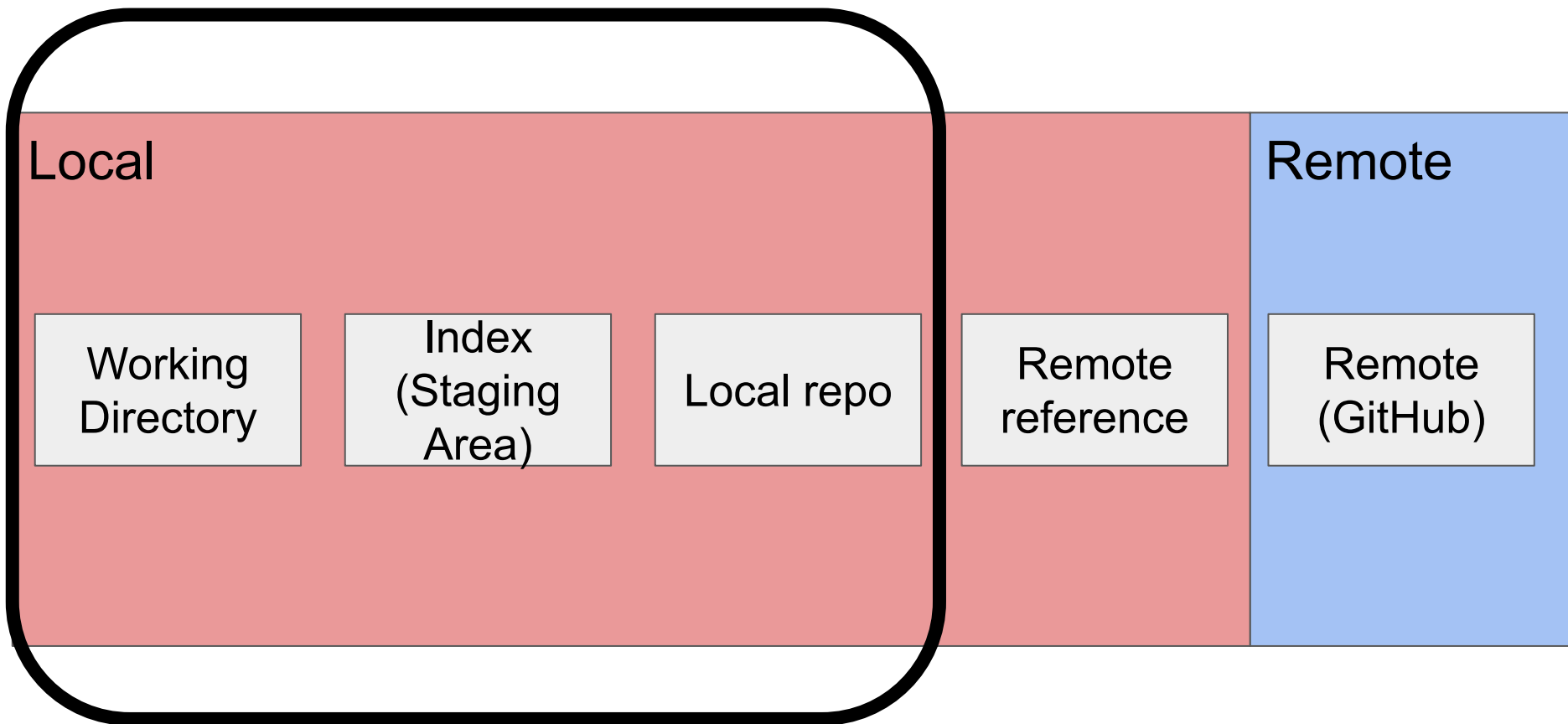
git - remotes

Adapted from materials by Dr. Carrier

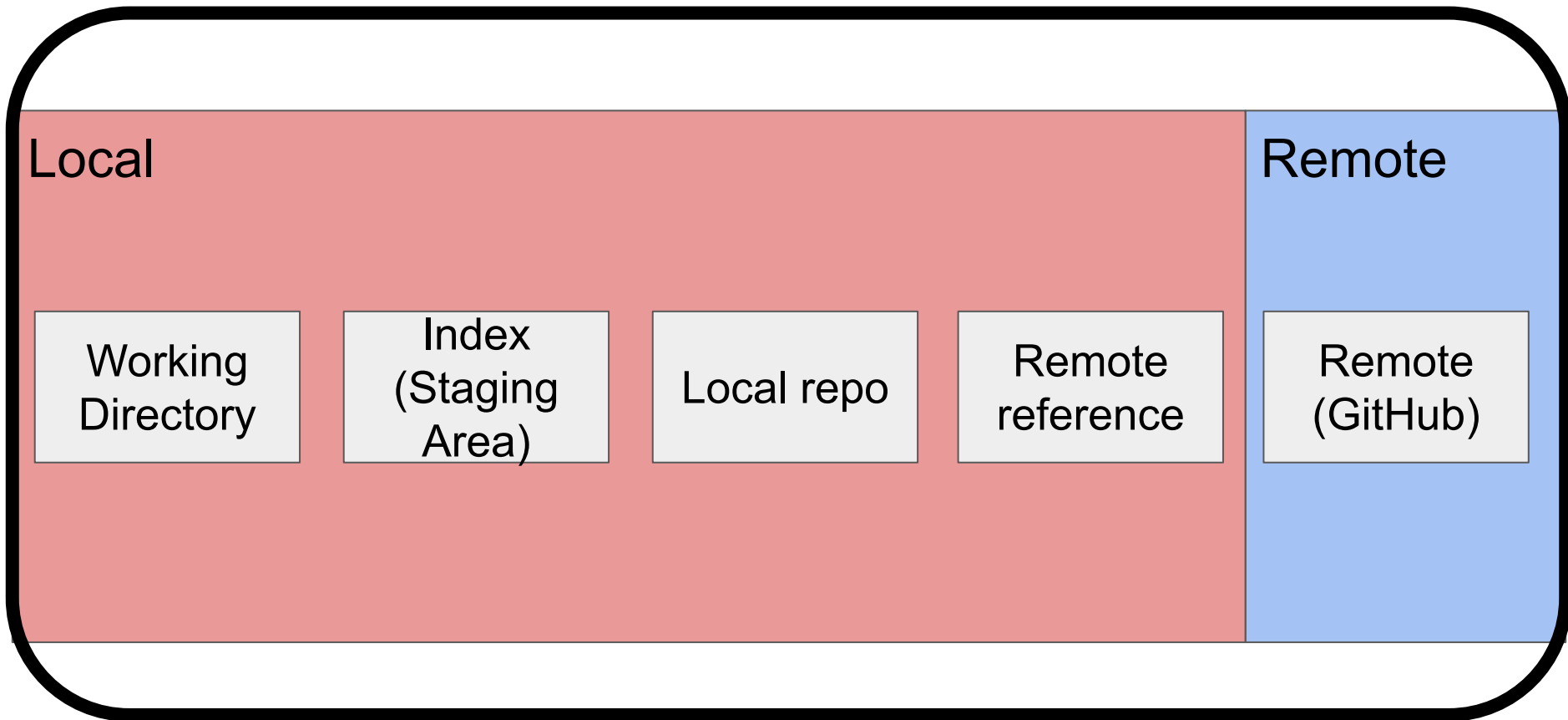


Where are we?

Working locally



Working locally + remotely



Where are we?

Where are we?

So far, we've only been working locally

Where are we?

So far, we've only been working locally

Now we want to connect to a *remote*

Where are we?

So far, we've only been working locally

Now we want to connect to a *remote*

Remote - A git repo stored elsewhere on a network

Where are we?

So far, we've only been working locally

Now we want to connect to a *remote*

Remote - A git repo stored elsewhere on a network

Here we'll use repos stored on GitHub



Where are we?

So far, we've only been working locally

Now we want to connect to a *remote*

Remote - A git repo stored elsewhere on a network

Here we'll use repos stored on GitHub

Other sites include GitLab, BitBucket, & Codeberg



Getting connected

Getting connected

Note: I generally recommend using SSH keys to interact with GitHub:

<https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account>

Getting connected

Getting connected

If you have a local repo you want to push to GitHub:

Getting connected

If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub

Getting connected

If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub
2. In your local repo, run:

```
git remote add origin remote_repo_url  
git push --set-upstream origin main
```

Getting connected

If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub
2. In your local repo, run:

```
git remote add origin remote_repo_url  
git push --set-upstream origin main
```

If you do *not* have a local repo:

Getting connected

If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub
2. In your local repo, run:

```
git remote add origin remote_repo_url  
git push --set-upstream origin main
```

If you do *not* have a local repo:

1. Grab the repo url off GitHub

Getting connected

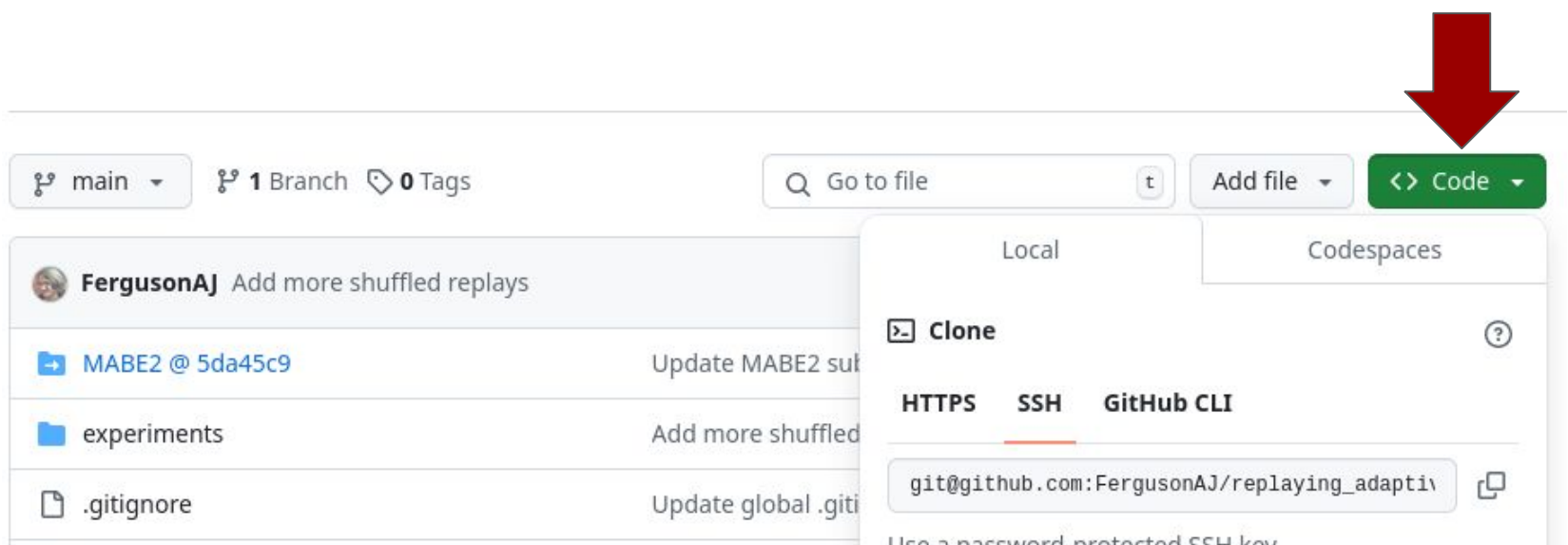
If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub
2. In your local repo, run:

```
git remote add origin remote_repo_url  
git push --set-upstream origin main
```

If you do *not* have a local repo:

1. Grab the repo url off GitHub



Getting connected

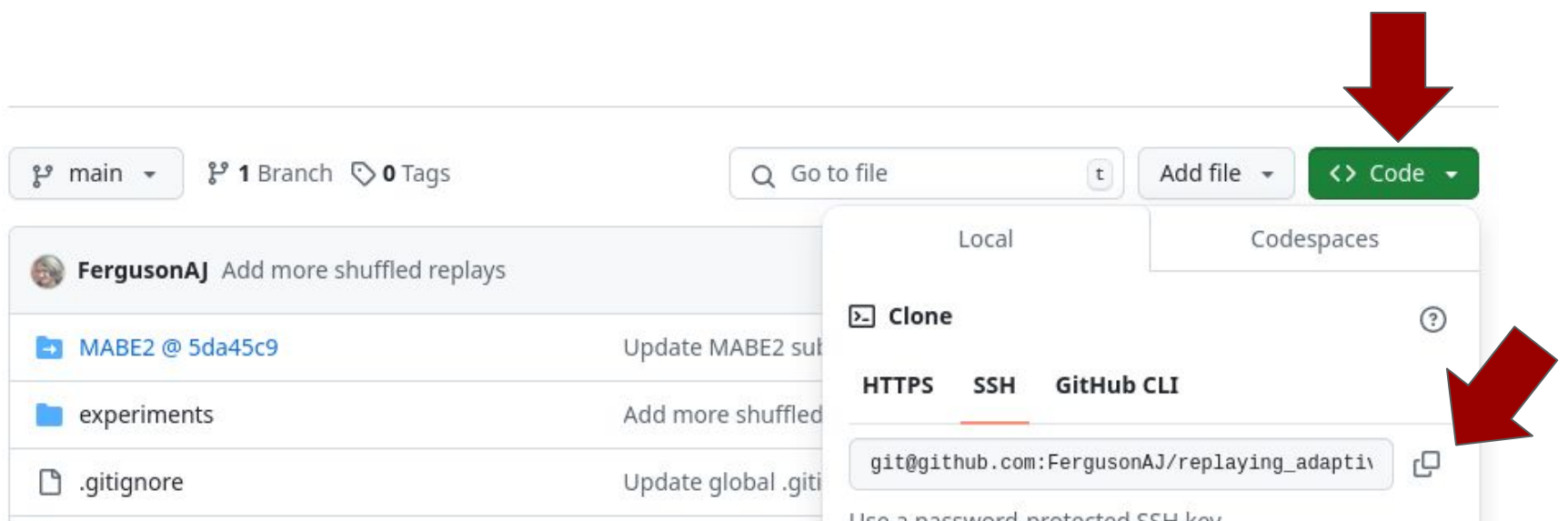
If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub
2. In your local repo, run:

```
git remote add origin remote_repo_url  
git push --set-upstream origin main
```

If you do *not* have a local repo:

1. Grab the repo url off GitHub



Getting connected

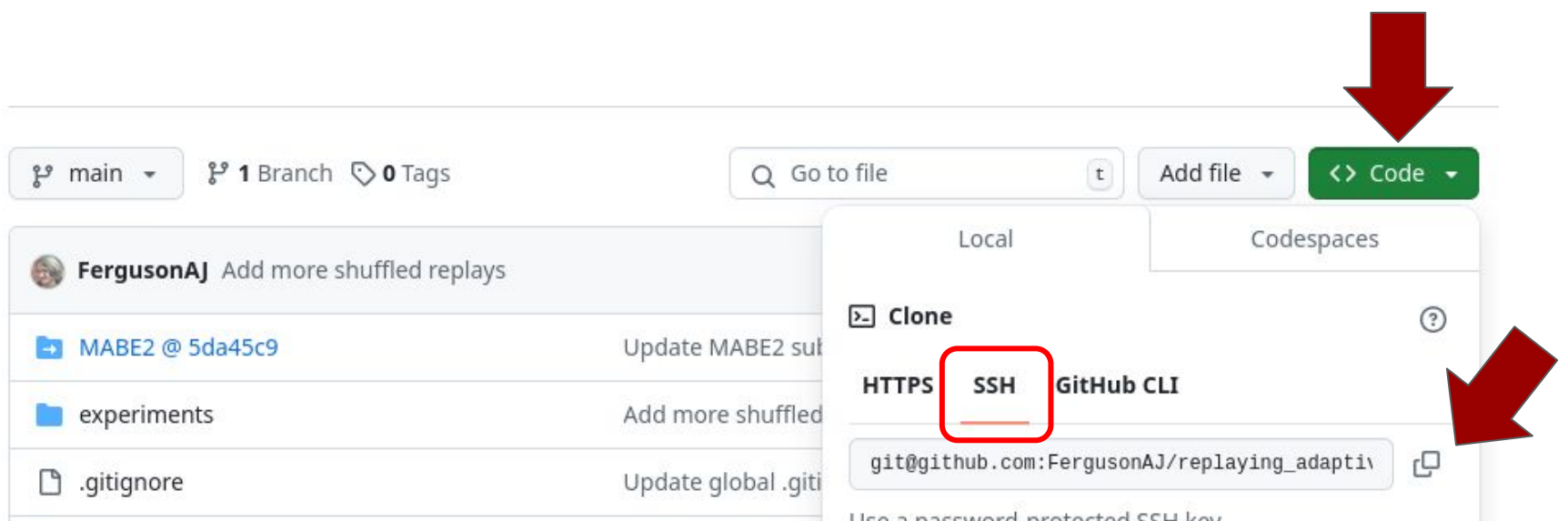
If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub
2. In your local repo, run:

```
git remote add origin remote_repo_url  
git push --set-upstream origin main
```

If you do *not* have a local repo:

1. Grab the repo url off GitHub



Getting connected

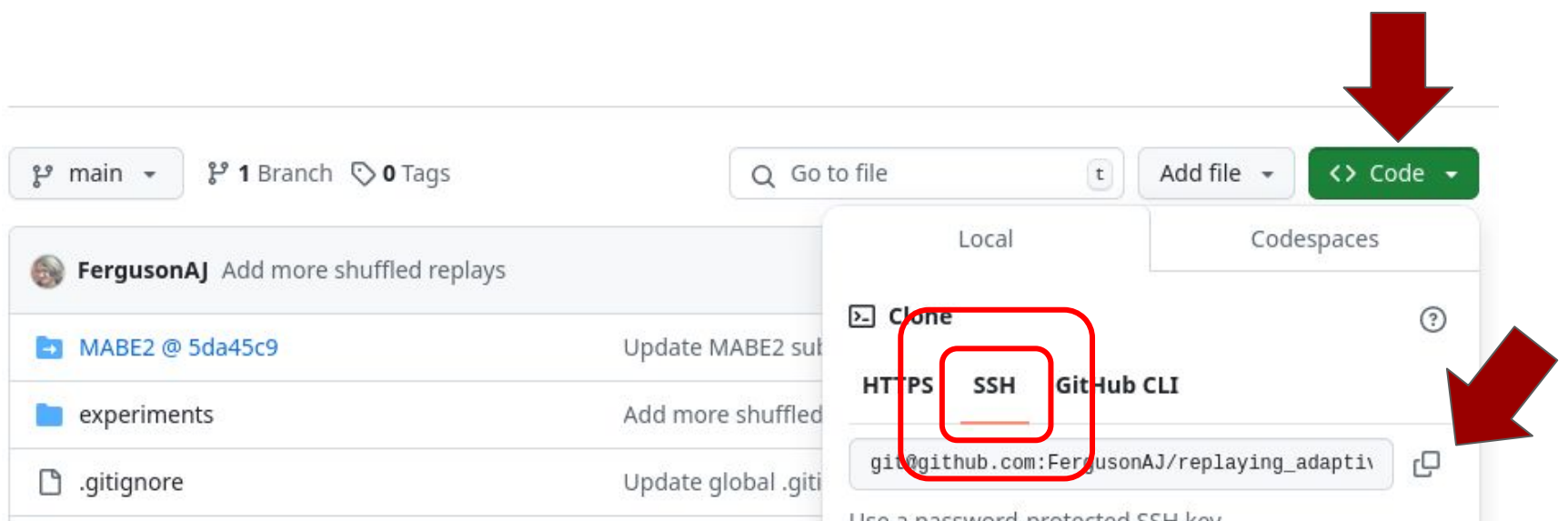
If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub
2. In your local repo, run:

```
git remote add origin remote_repo_url  
git push --set-upstream origin main
```

If you do *not* have a local repo:

1. Grab the repo url off GitHub



Getting connected

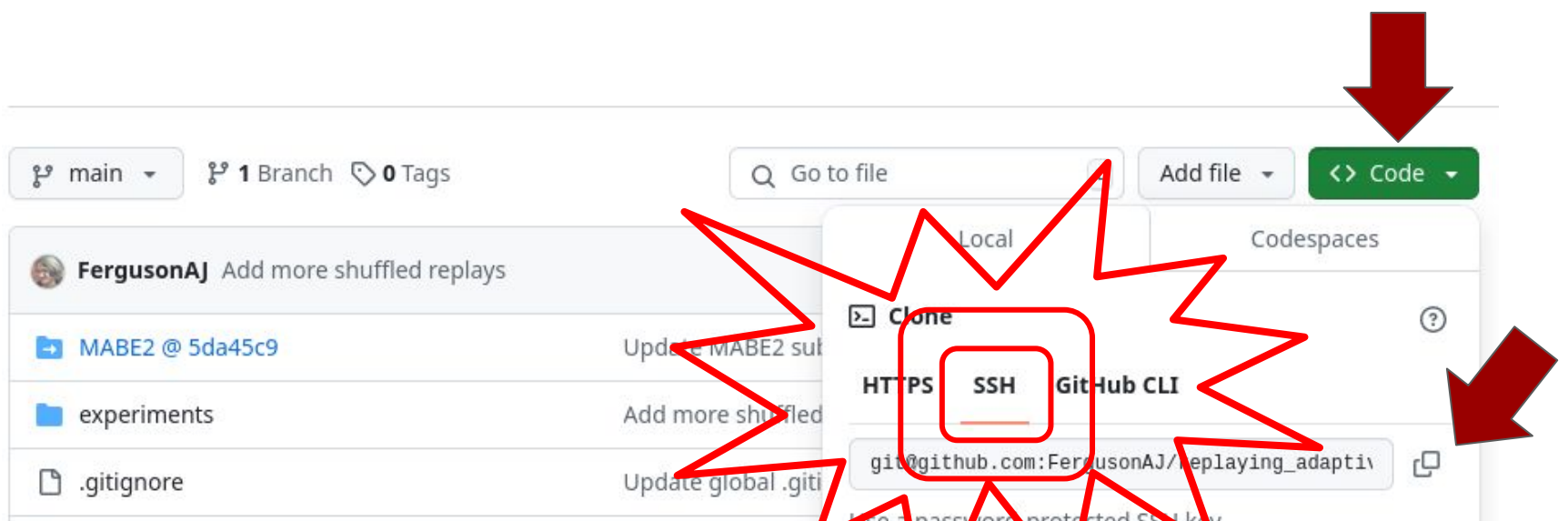
If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub
2. In your local repo, run:

```
git remote add origin remote_repo_url  
git push --set-upstream origin main
```

If you do *not* have a local repo:

1. Grab the repo url off GitHub



Getting connected

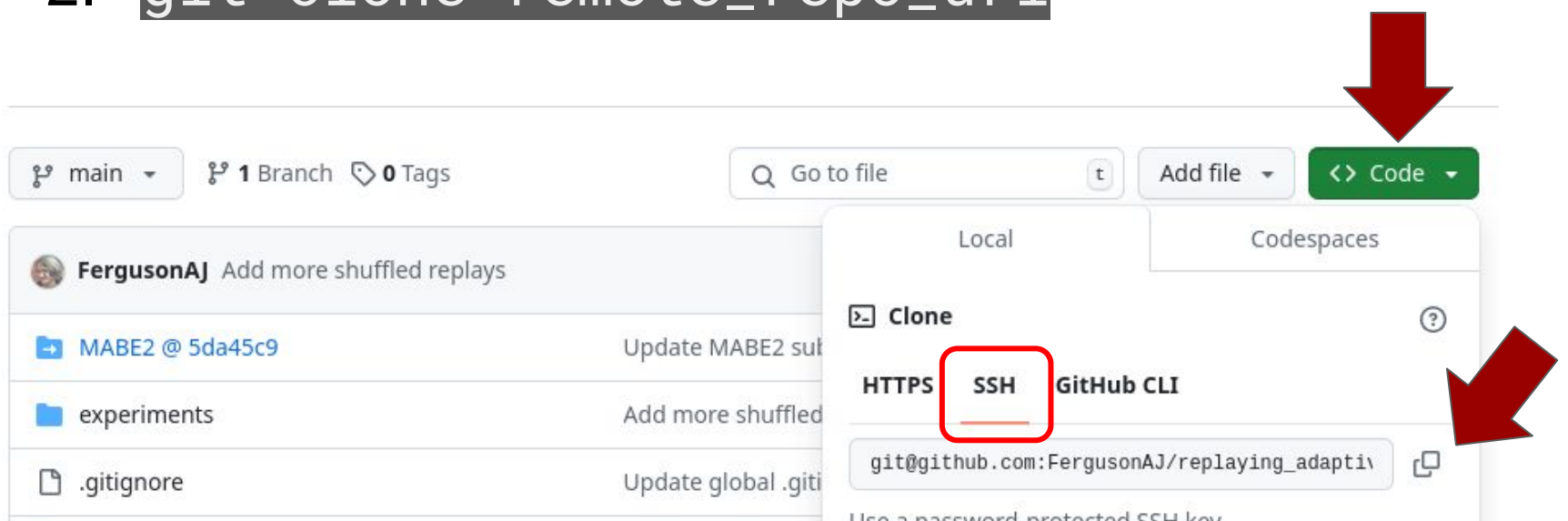
If you have a local repo you want to push to GitHub:

1. Create an empty repo on GitHub
2. In your local repo, run:

```
git remote add origin remote_repo_url  
git push --set-upstream origin main
```

If you do *not* have a local repo:

1. Grab the repo url off GitHub
2. `git clone remote_repo_url`



Workflow

Editing files, staging changes, and commits are the same as when working locally!

Workflow

Editing files, staging changes, and commits are the same as when working locally!

When you want to send your commits to remote:

Workflow

Editing files, staging changes, and commits are the same as when working locally!

When you want to send your commits to remote:

```
git push
```

Workflow

Editing files, staging changes, and commits are the same as when working locally!

When you want to send your commits to remote:

```
git push
```

```
git push remote branch (if needed)
```

Workflow

Editing files, staging changes, and commits are the same as when working locally!

When you want to send your commits to remote:

```
git push
```

```
git push remote branch
```

 (if needed)

When you want to download changes from remote:

Workflow

Editing files, staging changes, and commits are the same as when working locally!

When you want to send your commits to remote:

```
git push
```

```
git push remote branch (if needed)
```

When you want to download changes from remote:

```
git fetch
```

```
git merge
```

Workflow

Editing files, staging changes, and commits are the same as when working locally!

When you want to send your commits to remote:

```
git push
```

```
git push remote branch (if needed)
```

When you want to download changes from remote:

```
git fetch
```

```
git merge
```

Or

```
git pull
```

Warning!

Generally bad practice to rewrite history once it has been pushed to a remote

Warning!

Generally bad practice to rewrite history once it has been pushed to a remote

Generally, don't:

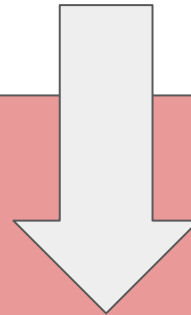
- Rebase
- Reset to an earlier commit
- Commit from detached HEAD
- Amend a commit that has been pushed

The final mystery



The final mystery

What's going on here?



Local

Working
Directory

Index
(Staging
Area)

Local repo

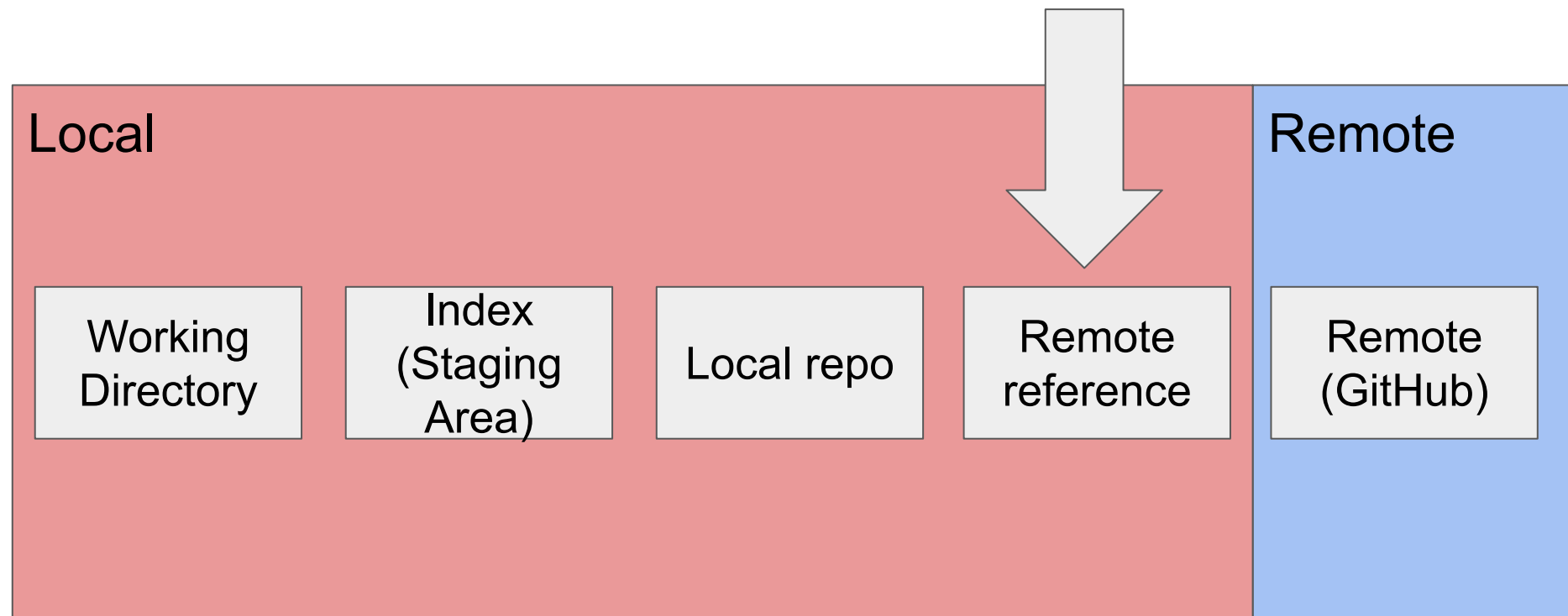
Remote
reference

Remote

Remote
(GitHub)

The final mystery

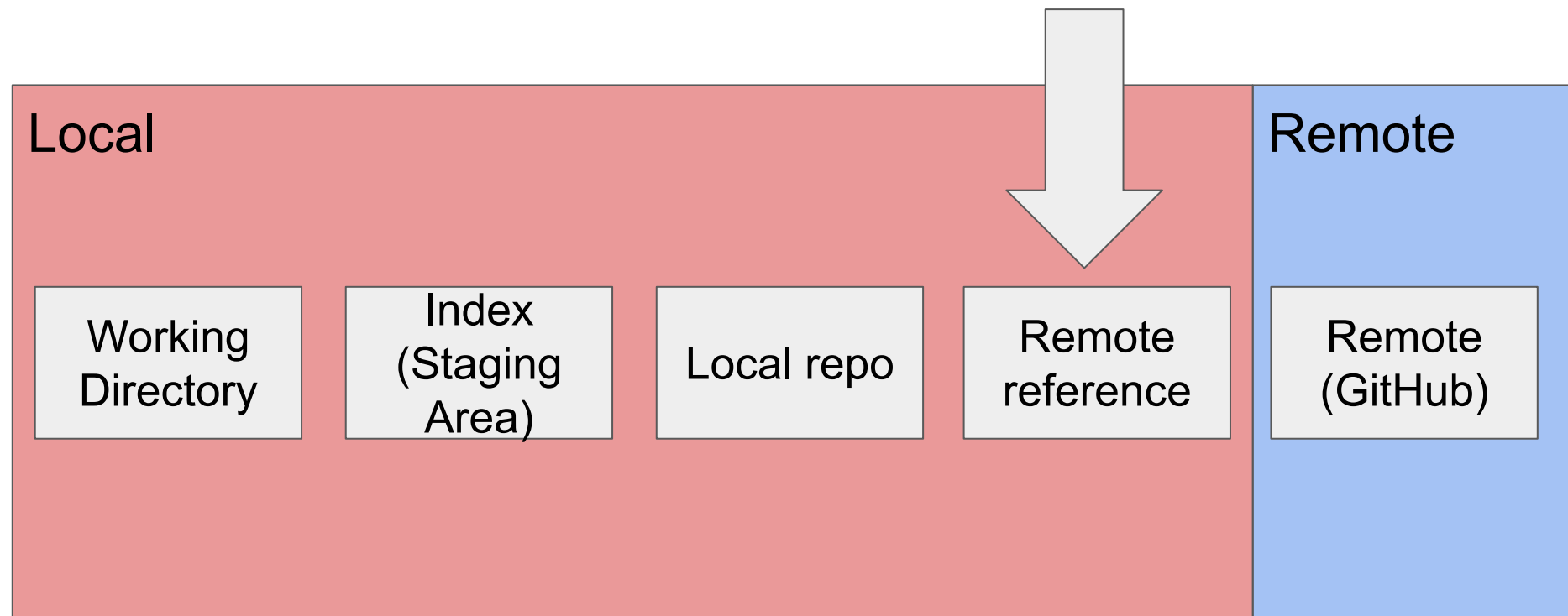
What's going on here?



Your local repo stores a copy of each branch on the remote

The final mystery

What's going on here?



Your local repo stores a copy of each branch on the remote

Thankfully, if we set our upstream correctly we do not need to interface with these much. :^)